

Rohan Butani

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EDUCATION

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

B.S. in Computer Science, B.S. in Applied Mathematics and Statistics | GPA: 4.00/4.00

Expected May 2028

- **Activities and Societies:** Marshal L. Salant Investment Team, Phi Delta Theta Fraternity, Quiz Bowl, SALIH

EXPERIENCE

Johns Hopkins University Data Science and AI Institute

Baltimore, MD

Research Assistant, Prof. Soufiane Hayou

September 2026 – Present

- Designing and evaluating multi-agent LLM systems to quantify how targeted agent adaptations propagate through group decision-making, information diffusion, and scaling dynamics under diverse orchestrated communication protocols.
- Leveraging the JHU-ARCH HPC cluster to create and run GPU-intensive multi-agent experiments at scale, benchmarking larger agent populations, model configurations, and communication structures through distributed inference and evaluation.

Johnson & Johnson

Cincinnati, OH

Applied AI Intern

August 2026 – Present

- Developing AI/ML solutions for MedTech R&D, applying statistical learning and data-driven methods to support medical device research, engineering workflows, and model-informed technical decision-making across R&D teams.
- Exploring translational applications focused on computational approaches to therapeutic discovery and development.

Franklin Templeton

Baltimore, MD

Investment Machine Learning Intern

June 2026 – August 2026

- Developed and trained a parallelized, agent-based financial document parsing and ingestion pipeline (Snowflake, Databricks, MLflow) to clean and process over 50,000 unstructured investment documents with >99.8% accuracy.
- Conducted alternative data research for catastrophe bond modeling using PySpark and Polars to evaluate meteorological, climate, and reinsurance-based signals related to insurance-linked securities risk/return behavior.
- Developed advanced time-series models using Snowflake, SQL, PyArrow, and PyTorch to forecast alpha decay and volatility dynamics for fixed income strategies, enabling systematic evaluation of signal persistence and risk.

Johns Hopkins University School of Medicine

Baltimore, MD

Research Assistant, Prof. Jeffrey Tornheim

September 2025 – February 2026

- Developed longitudinal machine learning and survival-analysis pipelines to model therapeutic response and outcomes in multidrug-resistant tuberculosis, identifying clinical and drug-related factors associated with efficacy and treatment failure.
- Built Python ETL pipelines to harmonize longitudinal clinical data into analysis-ready datasets for patient-level modeling.

The Wharton School Institute of AI and Analytics

Remote

Student Researcher, Lead and Corresponding Author

June 2025 – October 2025

- Scraped and cleaned NFL performance and health data with Python (requests, BeautifulSoup, Selenium, Pandas).
- Engineered demographic, travel, and workload features; trained and tuned classification models (scikit-learn, XGBoost, PyTorch (TabNet), TensorFlow) predicting injured reserve placement, processing over 20,000 records.

Georgia Institute of Technology School of Mathematics

Remote

Student Researcher, Lead and Corresponding Author

May 2024 – June 2025

- Engineered end-to-end machine learning pipelines in Python for predicting UCL reconstruction in baseball pitchers, achieving a novel 79.2% accuracy. Implemented feature importance visualization with SHAP and Matplotlib.
- Published peer-reviewed results as lead author in the *Proceedings of the International Conference on Healthcare Informatics*, indexed on IEEE Xplore. Conference travel funded by a \$3,500 National Science Foundation Grant.

RESEARCH PROJECTS

Sparse Autoencoder Hallucination Detection

August 2026 – Present

- Developing sparse autoencoder (SAE) probes for hallucination detection in large language models, localizing uncertainty-related representations across transformer layers using semantic-entropy labels on 7,382 prompts.
- Identifying predictive SAE features through statistical and sparse feature-selection methods, achieving ~0.82 AUROC with Gemma-Scope activations and analyzing feature contributions through decoder directions + logits.

LLM Benchmark Robustness & Ablation Analysis

May 2025 – Present

- Built automated Python pipelines using Hugging Face Transformers and OpenAI API to evaluate LLM artifact sensitivity through partial-input ablations on multiple-choice reading comprehension (MC-RC) benchmarks.
- Automated 100K+ GPU evaluations with batched inference, caching, and structured result logging using Pandas and JSON storage; work accepted to NeurIPS 2025 Workshops on LLM Evaluation and Efficient Reasoning.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, C, SQL, JavaScript, R, MATLAB, LaTeX
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, XGBoost, FastAPI, React, Pandas, Polars, PySpark, NumPy, PyArrow, MLflow, Hugging Face Transformers, Matplotlib, BeautifulSoup, Selenium, Requests, SciPy
- **Tools & Platforms:** Linux, Slurm, Git, GitHub, Docker, Kubernetes, Airflow, PostgreSQL, AWS, Snowflake, Databricks